CHE 201 Fall 2019 HW 2

Joey Wilczynski

TOTAL POINTS

72 / 100

QUESTION 1

1 Question 1 5 / 20

- **- 0 pts** Correct
- ✓ **- 2 pts Did not state if kinetic energy is increasing or decreasing**
- **- 1 pts** Did not state what would be the effect of friction on DeltaKE
- **- 2 pts** h = -1.5; thus DeltaPE = -5.88 J and Delta KE is 6.1 J
- **- 5 pts** Did not take work into consideration to calculate Delta KE. Change in Kinetic Energy = 6.1 J (increasing).
- **- 5 pts** Did not convert force into energy (Work)
- ✓ **- 3 pts Did not evaluate the effect of friction on DeltaKE**
- **- 1 pts** W = 0.05 * 4.39 = 0.219 J
- **- 2.5 pts** Accounted twice for the gravitational force. Delta KE is 6.1 J
- **- 2 pts** Kinetic energy is increasing
- ✓ **- 10 pts Did not calculate Delta KE**
- **- 1 pts** With friction, the change in kinetic energy (Delta KE) decreases - not the kinetic energy (KE).

QUESTION 2

2 Question 2 10 / 20

- **- 0 pts** Correct
- **- 1 pts** Work is negative
- **- 1 pts** W = -20000 J = -20 kJ
- **- 1 pts** Did not state the value used for Pf
- **- 5 pts** Did not show the solution for the integral
- ✓ **- 10 pts Wrong formula for work**
- **- 3 pts** Wrong unit. Work = -20 kJ
- **- 5 pts** Wrong solution for the integral. Work = -20 kJ
- **- 3 pts** Did not show how the integral results in -20 kJ
- **- 2 pts** Did not show the conversion from bar.m3 to kJ
- **- 1 pts** Missing the unit for the final volume
- **- 3 pts** Used wrong value for V2 in b). V2 = 0.0333 and W = -20 kJ
- **- 1 pts** The denominator in the work formula should be 1 - n
- **- 5 pts** Wrong calculation for P2V2 (= 29970 J) and P1V1 (= 10000 J). W = -20 kJ
- **- 2 pts** Wrong answer for final volume (= 0.0333 m3)
- **- 2 pts** wrong answer for Work (= -20000 J = -20 kJ)
- **- 1 pts** Integral solution should be -P2V2 + P1V1

QUESTION 3

3 Question 3 17 / 20

- **- 0 pts** Correct
- **- 2 pts** Stages 2 and 3 are inverted
- **- 4 pts** Wrong or missing the volume at state 2, in ft3
- **- 4 pts** Wrong or missing work calculation for process 1-2
- **- 4 pts** Wrong or missing work calculation for process 2-3. Work 2-3 = 0 Btu
- **- 4 pts** Wrong or missing work calculation for process 3-1
- **- 1 pts** Sketch missing labels, axis units and/or process arrows
- ✓ **- 1 pts Wrong sketch for process 1-2 (PV = constant is a curved line)**
- **- 3 pts** W1-2 and W3-1 are inverted
- **- 2 pts** Didn't show the integral calculation for Work 3-1
- **- 2.5 pts** Unit conversion to Btu is not clear
- **- 2.5 pts** Wrong or missing unit conversion to Btu
- ✓ **- 2 pts Wrong work formula for process 2-3**

- **1 pts** Wrong conversion from in2 to ft2
- **2 pts** Didn't show the integral calculation for Work

1-2

- **1 pts** Volume at state 2 should be in ft3
- **20 pts** Wrong answer

QUESTION 4

4 Question 4 **20 / 20**

✓ **- 0 pts** **Correct**

- **5 pts** calculation of "rate of conduction" was wrong
- **2 pts** Lb>= 11.22
- **10 pts** rate of conduction ?
- **2 pts** thickness of the wall?
- **5 pts** thickness of the wall calculation
- **20 pts** Question 4 missing

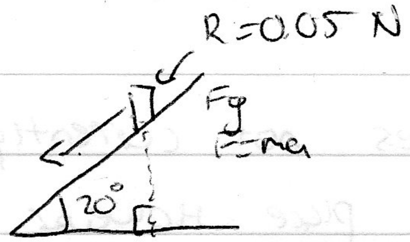
QUESTION 5

5 Question 5 **20 / 20**

✓ **- 0 pts** **Correct**

- **20 pts** Question 5 missing

#1.)



0.4kg

1.5m

$F = mg \cdot \sin\theta \Rightarrow (0.4)(9.8)\sin(20°) = 1.34N$

$\Delta KE = \int_{s_1}^{s_2} F\,ds \Rightarrow$

$\Delta KE =$

The increasing slant of the convayorbelt would make a difference in how much energy the car has, the higer the car ~~was~~ the higher the potential energy, and as the car goes down it transfers to keretic energy.

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- ✓ **- 10 pts Did not calculate Delta KE**
- **- 1 pts** With friction, the change in kinetic energy (Delta KE) decreases - not the kinetic energy (KE).

2.) $V_2 = \sqrt{\frac{P_1 \cdot V_1^2}{P_2}} \Rightarrow \sqrt{\frac{(1)(0.1)}{9}} = .0\overline{33}\ m^3$

$\boxed{A)\ .0\overline{33} m^3}$

$W \int P\,dV = \int \frac{c}{v^2}\,dv = c\left(\frac{-1}{V_2 - V_1}\right)$

$c = P_1 V_1^2 = P_2 V_2^2 = (900{,}000pa)(.033\ m^3) - \underline{(100{,}000pa)(0.1\ m^3)^2} \cdot \frac{1}{.03 - .1}$

$W = -19.97\ kj$

2 Question 2 **10 / 20**

- **- 0 pts** Correct
- **- 1 pts** Work is negative
- **- 1 pts** W = -20000 J = -20 kJ
- **- 1 pts** Did not state the value used for Pf
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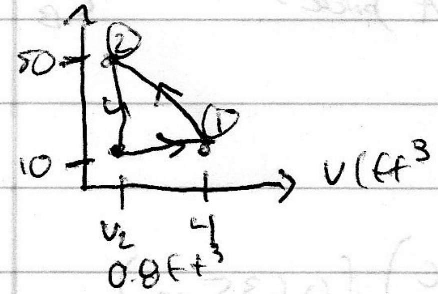
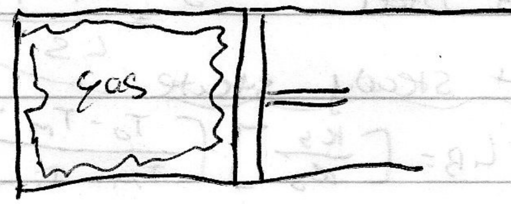
- **- 2 pts** Wrong answer for final volume (= 0.0333 m3)
- **- 2 pts** wrong answer for Work (= -20000 J = -20 kJ)
- **- 1 pts** Integral solution should be -P2V2 + P1V1

#3.)





known: $P = 10 \frac{lbf}{in^2} \times \frac{144 in}{ft^2} = \frac{1,140\ lbf}{ft^2}$

$P_2 = \frac{50}{in^2} \times \frac{144}{ft^2} = \frac{7200\ lbf}{ft^2}$

$V_1 = 4ft^3$

$V_2 = 0.8ft^3$

A.) $P_1V_1 = k$

$P_2V_2 = k$

$\frac{1440 lbf}{ft^2} \cdot 4^{ft^3} = k = 5760$

B.) 1-2: $\int_4^{0.8} \frac{dV}{V} = \ln(V)\Big|_4^{0.8} = \ln\left(\frac{0.8}{4}\right) \times 5760\ lbf \cdot$

$-9270.36\ lbf \cdot f \times \frac{9.4782 \times 10^{-21}}{0.73756}$

$\boxed{11.41\ BTU}$

2-3: $W \int_{0.8}^{0.8} \frac{dV}{V} = \ln(V)\Big|_{0.8}^{0.8} = \ln\left(\frac{0.8}{0.8}\right) \boxed{0\ BTU}$

3-1: $W = P(V_2 - V_1) = 1440 \frac{lbf}{ft^2} (4 - 0.8) ft^3 = 4608\ lbf \cdot ft$

$4608\ lbf \cdot ft \cdot \frac{9.4782 \times 10^{-4}}{0.73756} \frac{Btu}{lbf \cdot ft}$

$\boxed{5.92\ Btu}$

3 Question 3 **17 / 20**

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- **- 2 pts** Stages 2 and 3 are inverted
- **- 4 pts** Wrong or missing the volume at state 2, in ft3
- **- 4 pts** Wrong or missing work calculation for process 1-2
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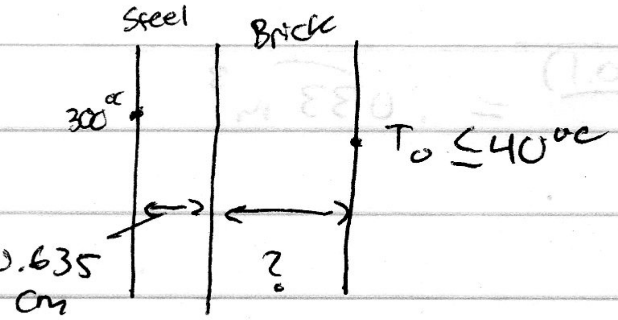
- ✓ **- 2 pts Wrong work formula for process 2-3**
- **- 1 pts** Wrong conversion from in2 to ft2
- **- 2 pts** Didn't show the integral calculation for Work 1-2
- **- 1 pts** Volume at state 2 should be in ft3
- **- 20 pts** Wrong answer



Model

A.) - Brick + steel system is closed system

- No work, At steady state, T changes linearly through each layer

Analysis: conductive heat transfer

$$\dot{Q} = -kA\frac{[t_2 - t_1]}{2}$$

$$\left(\frac{\dot{Q}}{A}\right)_{steel} = -k_s\frac{[t_m - t_i]}{L_S} = \left(\frac{\dot{Q}}{A}\right)_{brick} = \frac{-k_b[t_o - t_m]}{L_B}$$

At steady state

$$L_B = \left[\frac{k_s}{k_s}\right]\left[\frac{T_o - T_m}{T_m - T_i}\right] L_S$$

$$L_B \geq \left(\frac{0.7\ W/m.K}{15\ W/m.K}\right)\left(\frac{40 - 299.3°C}{-0.7°C}\right)(0.635\ cm)$$

$$\boxed{L_B \geq 11.22\ cm}$$

C.) $\left(\frac{\dot{Q}}{A}\right)_{steel} = \left(-15.1\ \frac{W}{m.K}\right)\left(\frac{299.3 - 300}{0.635cm}\right)$

$$\left(\frac{100cm}{1m}\right)\left(\frac{1kW}{1000\ W}\right)$$

$$\boxed{= 1.66\ kW/m^2}$$

4 Question 4 **20 / 20**

- ✓ **- 0 pts** **Correct**
- **- 5 pts** calculation of "rate of conduction" was wrong
- **- 2 pts** Lb>= 11.22
- **- 10 pts** rate of conduction ?
- **- 2 pts** thickness of the wall?
- **- 5 pts** thickness of the wall calculation
- **- 20 pts** Question 4 missing

#5) The state of illinois does not currently have battery regulations in place. However there are Federal requirements for battery recycling is that a company called called 2 Recycle helps battery and production manufactures fulfill recycling requirements in the U.S. There is also the mercury-containing and rechargeable battery act. The US federal law requires, with certain exceptions, used nickel cadium and lead batteries to be manages as universal waste. The universal waste rule prohibits handlers from disposing of wast batteries and they have to be recycled. Currently registration is working on cleaner battery technology, and illinois is the number 6 state that collects the most batteries by weight based on population

Source: call2recycle.org

☆ static at end based on 2018

5 Question 5 **20 / 20**

✓ **- 0 pts** **Correct**

- 20 pts Question 5 missing